

XPG Smart Soldering Station

User Manual

Version 1.0

High-performance intelligent soldering tool based on ESP32-S3 dual-core
microcontroller

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1. Product Overview

The XPG Smart Soldering Station is a high-performance intelligent soldering tool based on ESP32-S3, equipped with a 3.5-inch color touchscreen, supporting multiple heating element models, with PID precise temperature control, multiple safety protections, WiFi remote control, and other features.

1.1 Supported Heating Element Models

T12, C245, C210, C470 (auto-detection supported)

1.2 Main Features

- PID intelligent temperature control, fast heating, stable temperature
- 3.5-inch color touchscreen, 5 color themes
- Chinese/English bilingual interface
- WiFi connection, supports mobile/PC remote monitoring and control
- Multiple safety protections (overcurrent, overtemperature, undervoltage, tip change protection)
- Intelligent sleep and auto shutdown
- Buzzer sound feedback
- Custom startup logo
- Custom theme background
- Custom theme colors
- Multi-point temperature calibration
- OTA support

2. First Time Use

2.1 Power On

1. Connect power (supports 12~48V DC power supply, select appropriate voltage based on heating element model)
2. Install heating element (device auto-detects model)
3. Startup screen displays after power on

2.2 Initial Setup

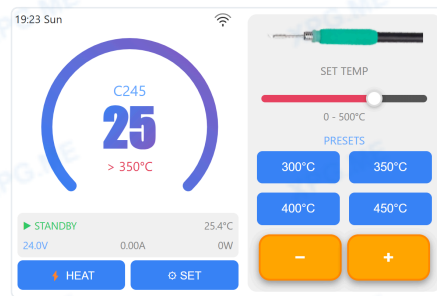
First boot requires the following initial settings:

1. Touchscreen calibration: Click the crosshairs appearing at the four corners of the screen in sequence, then click the center to confirm. Note: Touchscreen calibration is only for resistive touchscreens, capacitive touchscreen systems skip this step automatically
2. Language selection: A language selection dialog appears, choose "Chinese" or "English"

2.3 Basic Operation Flow

1. Click the "Heat" button on the main interface to start heating
2. Set target temperature via slider or +/- buttons
3. Wait for temperature to reach, buzzer indicates ready
4. Start soldering work
5. After use, click "Heat" button to turn off, or place on stand to automatically enter sleep mode

3. Main Interface



The main interface is the most frequently used interface for daily operation, containing temperature display, heating control, and parameter adjustment functions.

3.1 Heat Switch

- Click the heat button on the bottom left to turn heating on/off
- When on, button is highlighted; when off, returns to normal style
- After heating is turned on, PID controller automatically heats the tip to the set temperature

3.2 Temperature Setting

Three ways to set target temperature (range 0~500°C):

Slider	Drag the temperature slider on the right panel, continuous adjustment, real-time update
+/- Buttons	Click/hold [-][+] buttons, short press $\pm 10^{\circ}\text{C}$ each time, long press continuous adjustment
Presets	Click P1/P2/P3/P4 buttons to instantly switch to preset temperature

3.3 Real-time Information Display

- Current temperature: Large font display with PWM arc indicator
- Set temperature: Target temperature value
- Tip type: Currently detected heating element model
- Cold junction temperature: Ambient temperature at the handle
- Power voltage: Current input voltage
- Working current: Current heating current
- Heating power: Current output power
- Working status: Shows current status (Normal/Sleep/Standby/Protection, etc.)

3.4 Time, Day and WiFi Icon Display

When WiFi is successfully connected, the WiFi icon and time/day in the upper left corner will be displayed.

4. Settings Interface



Click the "Settings" button at the bottom of the main interface to enter the settings interface.

4.1 Heating Element Type

- Click the heating element button in the upper left corner to cycle: T12 → C245 → C210 → C470 → Auto Detect
- When "Auto Detect" is selected, the device automatically identifies the model based on the heating element ID pin
- After switching heating element type, corresponding calibration data and PID parameters will automatically switch

4.2 Sleep Temperature

- Click/hold [-][+] buttons to adjust sleep standby temperature
- Range: 0~300°C, step 10°C, default: 200°C

4.3 Sleep Timeout

- Click/hold [-][+] buttons to adjust the time to enter sleep after no operation
- Range: 1~30 minutes, default: 3 minutes

4.4 Preset Temperatures

- 4 preset temperature groups (P1~P4), each with [-][+] buttons
- Each click $\pm 10^{\circ}\text{C}$, range 0~500°C
- Default values: 300°C / 350°C / 400°C / 420°C

5. System Settings

5.1 PID Parameter Adjustment

PID parameters affect temperature control response speed and stability, generally no need to adjust.

Kp (Proportional)	Range 1.0~50.0, default 20.0, response speed
Ki (Integral)	Range 1.0~50.0, default 20.0, eliminate steady-state error
Kd (Derivative)	Range 1.0~50.0, default 20.0, suppress overshoot

Tip: If temperature is unstable or overshoot is severe, try reducing Ki and appropriately increasing Kd.

5.2 Protection Thresholds

Undervoltage Protection	Range 0~60.0V, default 6.0V, turns off heating when power voltage is below this value
Overcurrent Protection	Range 0~20.0A, default 10.0A, turns off heating when peak current exceeds this value

5.3 Other Options

Language Switch	Click "Language" button to cycle: Chinese / English
Sleep Mode	Click "Sleep Mode" button: Stand Sleep / Vibration Sleep
Boot Logo	Click switch to enable/disable boot logo display
Page Animation	Click switch to enable/disable page transition animations

6. Other Settings

The screenshot shows the 'Other' settings interface. On the left, there are three sliders: 'Backlight' set to 10, 'Volume' set to 5, and 'Shutdown' set to 30min. Below these is a power icon with the label 'Auto'. On the right side, there are three buttons: 'BG Color: White', 'Show Time', and 'IPS Screen'. At the bottom center are two buttons: 'BACK' and 'SAVE'.

6.1 Theme Style

Click the color button in the upper right corner to cycle through 6 themes:

White	Bright and clean, suitable for bright environments
Blue	Dark blue background, classic style
Black	Pure black background, darkroom friendly
Dark Gray	Dark gray background, soft and eye-friendly
Colorful	Dynamic gradient background, stunning visual effect
Custom	Customize background image and color scheme via Web settings panel

6.2 Backlight Brightness

- Drag slider to adjust screen brightness, range: 1 (darkest) ~ 10 (brightest)

6.3 Buzzer Volume

- Drag slider to adjust buzzer volume, range: 0 (mute) ~ 10 (maximum)

6.4 Auto Shutdown Time

- Drag slider to set auto shutdown time after sleep
- Range: 0~120 minutes, 0 means never shutdown, default: 30 minutes

6.5 Screen Saver Mode

Show Time	Black background displays large time and date, backlight at minimum
Turn Off Screen	Pure black screen with backlight off, minimum power consumption

6.6 Power Output Mode

Auto	Automatically calculates maximum power based on tip type and power voltage
Manual	User-defined PWM duty cycle (1~97%)
Maximum	Fixed 97% duty cycle, no power limit

Warning: Automatic power reduction at high voltage is to prevent heating element overload. For full power, select "Maximum" mode.

7. WiFi Settings



7.1 Scan and Connect

1. Click "Scan" button to search for surrounding WiFi hotspots
2. List shows each hotspot's name, signal strength, and encryption status
3. Click the hotspot to connect: open networks connect directly, encrypted networks require password input

7.2 Forget Network

- Click "Forget" button to disconnect current WiFi and clear saved password
- Need to re-enter password for next connection

7.3 Connection Status

Interface displays WiFi connection status in real-time: Not Connected / Connecting / Connected / Connection Failed. IP address is displayed when connected.

Tip: After WiFi connection, you can use mobile or PC browser to access the device IP address for remote control.

8. Temperature Calibration

Temperature	ADC Value	Adjustment
100°C	0	- +
200°C	0	- +
300°C	0	- +
400°C	0	- +
500°C	0	- +
600°C	0	- +

V Cal: 0.0V - +

Room Cal: 25.0°C - +

BACK SAVE

Temperature calibration is used to correct heating element temperature reading deviation, ensuring displayed temperature matches actual temperature.

Warning: Calibration requires an external reference thermometer. Incorrect calibration data will cause abnormal temperature control, please operate carefully.

8.1 Calibration Steps

1. Enter settings interface, click "Calibrate" button
2. Prepare a reliable external thermometer
3. Select the temperature point to calibrate (100/200/300/400/500/600°C)
4. Heat the iron to the corresponding temperature, wait for temperature to stabilize
5. Click that temperature point button to read current ADC value
6. Use [-][+] buttons to fine-tune ADC value until displayed temperature matches external thermometer reading
7. Repeat steps 3~6 to complete all needed calibration points
8. Click "Save" to save calibration data

8.2 Voltage Calibration

- Use [-][+] buttons to adjust voltage calibration coefficient
- Interface displays current voltage reading in real-time, adjust after comparing with multimeter
- Range: 2~400

8.3 Room Temperature Calibration (Cold Junction Compensation Offset)

- Use [-][+] buttons to adjust cold junction temperature offset
- Interface displays cold junction temperature reading in real-time, adjust after comparing with actual room temperature
- Range: -100~100 (unit 0.1°C)

9. Sleep & Screen Saver

9.1 Sleep Mode

The device supports two sleep trigger methods:

Stand Sleep	Triggered when iron is placed on stand, wake up when picked up from stand
Vibration Sleep	Triggered after no vibration for set time, wake up when iron is picked up or touched

During sleep, the tip maintains sleep standby temperature (default 200°C), saving energy while allowing quick return to working temperature.

9.2 Auto Shutdown & Screen Saver

- After sleep exceeds the set auto shutdown time, device enters screen saver mode
- Heating is completely off in screen saver mode

Show Time	Black background, large time+date, touch screen to exit
Turn Off Screen	Pure black screen, touch screen to exit

Warning: After exiting screen saver, enters normal working mode, but heating is off by default, needs to be manually turned on.

10. Safety Protection

The device has built-in multiple safety protection mechanisms to ensure safe operation.

10.1 Overcurrent Protection

Trigger Condition	Peak current exceeds set threshold (default 10A)
Trigger Behavior	Immediately turns off heating, status bar shows red "Overcurrent Protection" warning
Recovery Method	Requires device restart

10.2 Overtemperature Protection

Trigger Condition	Tip temperature exceeds 550°C
Trigger Behavior	Immediately turns off heating, buzzer alarms, status bar shows red warning
Recovery Method	Automatically recovers when temperature drops below 540°C

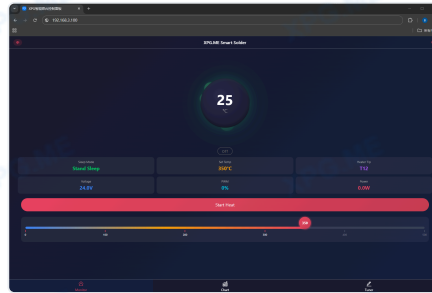
10.3 Undervoltage Protection

Trigger Condition	Power voltage below set threshold (default 6.0V)
Trigger Behavior	Turns off heating, buzzer alarms, status bar shows red warning
Recovery Method	Automatically clears when voltage recovers

10.4 Tip Change Protection

Trigger Condition	Tip insertion/removal detected
Trigger Behavior	Turns off heating, buzzer indicates, status bar shows orange warning
Recovery Method	Automatically recovers after 5 seconds

11. Remote Control (Web Control Panel)



After the device connects to WiFi, you can access the device IP address via mobile or PC browser for remote monitoring and control. The Web control panel provides three main pages: Monitor page, Temperature curve page, PID tuning page.

11.1 Access Method

1. Ensure device is connected to WiFi (WiFi signal icon displayed in upper left corner of main interface)
2. View device IP address in Settings → WiFi interface
3. Enter that IP address in mobile/PC browser on the same LAN
4. Language can be switched at the top of the page (Chinese/English), settings button in upper right corner for detailed settings

11.2 Monitor Page

- Temperature ball animation: Intuitively displays current temperature, color changes with temperature
- Status badge: Shows current working status (Normal, Sleep, Screen Saver, Overtemperature Protection, etc.)
- Status grid: Shows sleep mode, set temperature, tip type, voltage, PWM duty cycle, power
- Heat switch: Remote turn on/off heating, can remotely exit screen saver mode
- Temperature slider: Drag to set target temperature (0~500°C), with scale marks

11.3 Temperature Curve Page

- Top status card: Shows current temperature, set temperature, tip type
- Temperature curve: Dual-axis line chart drawn with Chart.js
- Fullscreen mode: Click fullscreen button for large display
- Clear data: Click clear button to reset curve history data

11.4 PID Tuning Page

- Target temperature setting: Set target temperature for step test
- Start/Stop test: Start or stop step response test
- PID parameter adjustment: Real-time adjustment of Kp, Ki, Kd parameters
- Step response curve: Real-time plotting of temperature response process
- Response metrics calculation: Automatically calculates rise time, overshoot, settling time, steady-state error
- History comparison: Save multiple test results to compare different PID parameter effects

11.5 Settings Panel

Click settings button in upper right corner to enter detailed settings panel, containing the following groups:

- Tip selection: T12/C245/C210/C470/Auto Detect
- Preset temperatures: 4 preset temperature adjustments
- Sleep settings: Sleep mode, sleep temperature, sleep timeout, auto shutdown time, screen saver mode
- Display & Sound: Language, backlight brightness, volume, color scheme
- Power mode: Auto/Manual/Maximum, PWM duty cycle can be adjusted in Manual mode
- Protection settings: Overcurrent protection threshold, undervoltage protection threshold
- Calibration: Voltage calibration, room temperature calibration, temperature ADC calibration (7-point calibration)
- PID parameters: Kp, Ki, Kd adjustment
- System settings: Boot logo, page animation, factory reset
- Firmware upgrade: Application firmware OTA, Web file OTA
- Custom logo: Upload/delete boot logo image

11.6 Custom Theme Colors

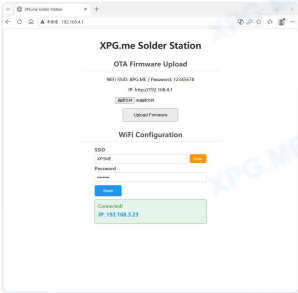
Customize LCD theme style via Web settings panel to make the device interface more personalized. Supports custom background images and color schemes.

Tip: Custom theme colors make your device unique, you can adjust interface style according to personal preference or working environment.

11.7 WebSocket Real-time Push

After browser connects, device pushes real-time status data every 100ms for dynamic monitoring. Supports up to 4 simultaneous client connections.

12. Firmware Upgrade (Factory Mode)



Factory mode is used for firmware upgrade and WiFi pre-configuration.

12.1 Enter Factory Mode

Main Program OTA Upgrade	Hold motherboard OTA button then connect power, manually enter Factory mode
No Valid Firmware	Device automatically enters (no manual operation needed)

12.2 Factory Mode Operation

After entering Factory mode, screen displays OTA upgrade guide interface:

1. Connect WiFi hotspot: Use mobile or PC to search and connect WiFi: SSID: XPG.ME, password: 12345678
2. Open browser: Visit <http://192.168.4.1>
3. Upload firmware: Click file selection button, select .bin firmware file, click "Upload Firmware" button
4. Wait for progress bar to complete, device automatically restarts, new firmware takes effect

13. WiFi Config without Screen (Factory Mode)

WiFi network can be pre-configured in Factory mode, device will automatically connect after restart.

13.1 Enter Factory Mode

Hold motherboard OTA button then connect power, buzzer beeps once to enter Factory mode.

13.2 WiFi Config Operation

After entering Factory mode, screen displays WiFi configuration guide interface:

1. Connect WiFi hotspot: Use mobile or PC to search and connect WiFi: SSID: XPG.ME, password: 12345678
2. Open browser: Visit <http://192.168.4.1>
3. Scan WiFi: Click "Scan" button to scan available WiFi networks
4. Select network: Select WiFi network to connect from the list
5. Enter password: Enter WiFi password in password field
6. Save configuration: Click "Save" button to save WiFi configuration
7. Device automatically restarts, will automatically connect to configured WiFi network after restart

Tip: After WiFi connection succeeds, obtained IP address will be displayed, visit that IP address in browser to access WEB interface for remote monitoring and control.

14. Factory Reset

Three ways to perform factory reset:

System Settings Interface	Drag "Factory Reset" slider to 100%, must drag to bottom to prevent accidental operation
Web API	Visit device webpage, click factory reset, remote operation
Hardware Button	Long press BOOT button for 5 seconds, BOOT button on back of motherboard

After factory reset:

- All user settings restore to default values
- Calibration data restores to default values
- Custom boot logo is deleted
- Custom theme background image is deleted
- Custom theme colors restore to default values
- Device automatically restarts

15. Troubleshooting

Problem	Solution
Displays 25°C after boot, not actual residual temperature	ADC warmup not complete, wait a few seconds for auto update
Temperature cannot reach set value	Power mode limit or insufficient power voltage
Severe temperature overshoot	PID parameters improper, try reducing Ki value, appropriately increase Kd value
Shows "Undervoltage Protection"	Power voltage too low, check power connection and voltage
Shows "Overcurrent Protection"	Abnormal current, check if heating element is shorted, restart device
Shows "Overtemperature Protection"	Temperature exceeds 550°C, wait for temperature to drop below 540°C for auto recovery
Shows "Tip Change Protection"	Tip change detected, wait 5 seconds for auto recovery
WiFi cannot connect	Password error or weak signal, re-enter password, move closer to router
Touch inaccurate	Touchscreen not calibrated, enter system settings to recalibrate touchscreen
Heating on but temperature doesn't rise	Heating element not installed or damaged, check heating element connection
Buzzer doesn't sound	Volume set to 0, increase buzzer volume in other settings
Shows room temperature but doesn't heat up	Check wiring fault, confirm heating element is good
No temperature display, heating element glowing red	Handle wiring error, check wiring according to wiring diagram

16. Technical Specifications

Controller	ESP32-S3 dual-core 240MHz
Display	3.5-inch ILI9488 color touchscreen, 320×480 resolution
Supported Heating Elements	T12 / C245 / C210 / C470
Temperature Range	0~500°C
Temperature Control	PID intelligent temperature control
Power Voltage	12~48V DC (select based on heating element model)
WiFi	802.11 b/g/n
Interfaces	Handle interface / Sleep interface / Tip change protection interface / Power interface
Protection Features	Overcurrent / Overtemperature / Undervoltage / Tip change protection
Language	Chinese / English

17. Safety Precautions

Warning: Please read and follow the following safety precautions carefully, otherwise it may cause personal injury or device damage!

17.1 High Temperature Burns

- Soldering tip can reach over 450°C during operation, contact with skin will cause severe burns
- Do not touch the tip or nearby metal parts with your hands
- When resting, please place the iron on the stand, never leave it randomly

17.2 Fire Safety

- Do not use this device near flammable materials
- Work area should be away from paper, cloth, solvents and other flammable items
- Ensure work surface is heat-resistant material

17.3 Long Time Idle

- When not using for a long time, please turn off power or unplug power adapter
- Although the device has sleep and screen saver functions, maintaining temperature for a long time will still accelerate tip oxidation

17.4 Tip Replacement

- Please replace tip after it has cooled down
- Hot plugging may cause burns and is not good for tip lifespan
- After replacement, system automatically enters 5-second tip change protection period, no heating during this time

17.5 Network Security

- WiFi password is stored in device NVS, not transmitted over network
- Web control panel has no built-in authentication mechanism, please use in trusted network environment
- Recommend not using Web control function in public WiFi networks

— End of Document —

Hardware is regularly maintained and updated. If there are discrepancies between the document and the actual product, please contact technical support or visit official website XPG.ME for latest information.

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